

|                |                                         |
|----------------|-----------------------------------------|
| Report Period  | Sep 15, 2026 10:15 - Sep 15, 2026 10:20 |
| Generated At   | 2026-09-25 14:18:26                     |
| Classification | ML: Isolation Forest + Random Forest    |

## 1. Executive Summary

|                         |        | Traffic Summary        |  | Count      |  |
|-------------------------|--------|------------------------|--|------------|--|
| Metric                  | Value  | Total Flows Observed   |  | 14893      |  |
| Total Threats Mitigated | 19     | Total Packets Analyzed |  | 5191528237 |  |
| High Priority           | 4      | Malicious Packets      |  | 5162954449 |  |
| Low Priority            | 15     | True Negatives Passed  |  | 10183      |  |
| Manual Releases         | 0      |                        |  |            |  |
| Manual Blocks           | 0      | Attack Vector          |  | Count      |  |
|                         |        | ICMP Flood             |  | 0          |  |
| ML Metrics              | Value  | SYN Flood              |  | 13         |  |
| False Positives         | 8      | UDP Flood              |  | 2          |  |
| FP Rate                 | 0.01%  | Uncertain              |  | 4          |  |
| IF Precision            | 99.98% |                        |  |            |  |
| IF Recall               | 93.80% | Action                 |  | Count      |  |
| IF F1-Score             | 96.79% | Quarantined            |  | 4          |  |
| IF Accuracy             | 97.91% | Time Ban               |  | 4          |  |
|                         |        | Blackhole              |  | 0          |  |
|                         |        | Released               |  | 2          |  |

## 2. Performance Benchmark

### 2a. Isolation Forest: Anomaly Detection

| Metric                    | Value  | Description                                          |
|---------------------------|--------|------------------------------------------------------|
| Precision                 | 99.98% | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | 93.80% | Share of actual attacks successfully flagged         |
| F1-Score                  | 96.79% | Balanced measure combining Precision and Recall      |
| Accuracy                  | 97.91% | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | 0.01%  | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | 6.20%  | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | 93.80% | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | 99.99% | Normal traffic correctly identified as safe          |

|                | Predicted: Attack | Predicted: Normal |
|----------------|-------------------|-------------------|
| Actual: Attack | TP 4709           | FN 311            |
| Actual: Normal | FP 1              | TN 9872           |

### 2b. Random Forest: Attack Classification

| Metric                    | Value   | Description                                          |
|---------------------------|---------|------------------------------------------------------|
| Precision                 | 100.00% | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | 100.00% | Share of actual attacks successfully flagged         |
| F1-Score                  | 100.00% | Balanced measure combining Precision and Recall      |
| Accuracy                  | 100.00% | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | 0.00%   | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | 0.00%   | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | 100.00% | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | 100.00% | Normal traffic correctly identified as safe          |

|           | Predicted: SYN | Predicted: ICMP | Predicted: UDP |
|-----------|----------------|-----------------|----------------|
| Act: SYN  | 4709           | 0               | 0              |
| Act: ICMP | 0              | 0               | 0              |
| Act: UDP  | 0              | 0               | 0              |

### 2c. Hybrid Cascaded

| Metric                    | Value  | Description                                          |
|---------------------------|--------|------------------------------------------------------|
| Precision                 | 99.98% | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | 93.80% | Share of actual attacks successfully flagged         |
| F1-Score                  | 96.79% | Balanced measure combining Precision and Recall      |
| Accuracy                  | 97.91% | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | 0.01%  | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | 6.20%  | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | 93.80% | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | 99.99% | Normal traffic correctly identified as safe          |

|                | Predicted: Attack | Predicted: Normal |
|----------------|-------------------|-------------------|
| Actual: Attack | TP 4709           | FN 311            |
| Actual: Normal | FP 1              | TN 9872           |

2d. Response Latency

| Metric                   | Value    | Description                                                             |
|--------------------------|----------|-------------------------------------------------------------------------|
| Detection Time           | 331.6 ms | Average time taken to detect an attack                                  |
| Mitigation Response Time | 8.8 ms   | Interval between the anomaly flag and the FlowMod blocking rule install |

2e. Controller Resource Overhead

| Metric                         | Value | Description                                                           |
|--------------------------------|-------|-----------------------------------------------------------------------|
| Controller CPU (Baseline)      | 1.06% | Ryu controller CPU usage under normal traffic                         |
| Controller CPU (Active Attack) | 7.14% | Ryu controller average CPU usage during a DDoS simulation             |
| Controller CPU (Mitigation)    | 4.74% | Ryu controller CPU usage during simultaneous detection and mitigation |

3. Offences Summary

| Source IP | Sessions | Max Offences | First Seen          | Last Seen           |
|-----------|----------|--------------|---------------------|---------------------|
| 10.0.0.19 | 3        | 3            | 2026-09-15 10:17:42 | 2026-09-15 10:19:03 |
| 10.0.0.16 | 2        | 2            | 2026-09-15 10:17:40 | 2026-09-15 10:18:50 |
| 10.0.0.17 | 2        | 2            | 2026-09-15 10:17:40 | 2026-09-15 10:18:50 |
| 10.0.0.18 | 2        | 2            | 2026-09-15 10:17:42 | 2026-09-15 10:18:52 |
| 10.0.0.6  | 1        | 1            | 2026-09-15 10:15:28 | 2026-09-15 10:16:29 |

#### 4. Chronological Mitigation Log

| Timestamp           | Source IP | Class | Vector    | Confidence | Priority | Action        |
|---------------------|-----------|-------|-----------|------------|----------|---------------|
| 2026-09-15 10:15:28 | 10.0.0.6  | DDoS  | UDP Flood | 92.4%      | High     | Time Ban (1m) |
| 2026-09-15 10:16:29 | 10.0.0.6  | DDoS  | UDP Flood | 92.4%      | High     | Released      |
| 2026-09-15 10:17:40 | 10.0.0.16 | DDoS  | Uncertain | -          | Low      | Quarantined   |
| 2026-09-15 10:17:40 | 10.0.0.17 | DDoS  | Uncertain | -          | Low      | Quarantined   |
| 2026-09-15 10:17:42 | 10.0.0.18 | DDoS  | Uncertain | -          | Low      | Quarantined   |
| 2026-09-15 10:17:42 | 10.0.0.19 | DDoS  | Uncertain | -          | Low      | Quarantined   |
| 2026-09-15 10:17:50 | 10.0.0.16 | DDoS  | SYN Flood | 92.5%      | Critical | Time Ban (1m) |
| 2026-09-15 10:17:50 | 10.0.0.17 | DDoS  | SYN Flood | 92.1%      | Critical | Time Ban (1m) |
| 2026-09-15 10:17:51 | 10.0.0.17 | DDoS  | SYN Flood | 92.1%      | Critical | Time Ban      |
| 2026-09-15 10:17:51 | 10.0.0.16 | DDoS  | SYN Flood | 92.5%      | Critical | Time Ban      |
| 2026-09-15 10:17:52 | 10.0.0.18 | DDoS  | SYN Flood | 90.9%      | Critical | Time Ban (1m) |
| 2026-09-15 10:17:52 | 10.0.0.19 | DDoS  | SYN Flood | 91.7%      | Critical | Time Ban (1m) |
| 2026-09-15 10:17:52 | 10.0.0.18 | DDoS  | SYN Flood | 90.9%      | Critical | Time Ban      |
| 2026-09-15 10:17:53 | 10.0.0.19 | DDoS  | SYN Flood | 91.7%      | Critical | Time Ban      |
| 2026-09-15 10:18:50 | 10.0.0.16 | DDoS  | SYN Flood | 92.5%      | Critical | Time Ban (2m) |
| 2026-09-15 10:18:50 | 10.0.0.17 | DDoS  | SYN Flood | 92.1%      | Critical | Time Ban (2m) |
| 2026-09-15 10:18:52 | 10.0.0.18 | DDoS  | SYN Flood | 93.3%      | High     | Time Ban (2m) |
| 2026-09-15 10:18:52 | 10.0.0.19 | DDoS  | SYN Flood | 93.4%      | High     | Released      |
| 2026-09-15 10:19:03 | 10.0.0.19 | DDoS  | SYN Flood | 93.4%      | Critical | Time Ban (2m) |

#### 5. IP Attack History (Completed Sessions)

| Source IP | Vector    | IF Score | Conf  | Priority | Phase   | Offences | Duration | Unblocked At        | Reason                                    |
|-----------|-----------|----------|-------|----------|---------|----------|----------|---------------------|-------------------------------------------|
| 10.0.0.19 | SYN Flood | 0.6537   | 93.4% | Critical | Phase 2 | 3        | 10s      | 2026-09-15 10:19:03 | Attack persisted - escalating to time ban |
| 10.0.0.19 | SYN Flood | 0.5913   | 93.4% | High     | Phase 2 | 1        | 1m 10s   | 2026-09-15 10:18:52 | Ban Expired                               |
| 10.0.0.18 | SYN Flood | 0.5775   | 93.3% | High     | Phase 2 | 2        | 1m 10s   | 2026-09-15 10:18:52 | Attack persisted - escalating to time ban |
| 10.0.0.17 | SYN Flood | 0.6430   | 92.1% | Critical | Phase 2 | 2        | 1m 10s   | 2026-09-15 10:18:50 | Attack persisted - escalating to time ban |
| 10.0.0.16 | SYN Flood | 0.6387   | 92.5% | Critical | Phase 2 | 2        | 1m 10s   | 2026-09-15 10:18:50 | Attack persisted - escalating to time ban |
| 10.0.0.19 | SYN Flood | 0.6521   | 91.7% | Critical | Phase 2 | 1        | 10s      | 2026-09-15 10:17:52 | Attack persisted - escalating to time ban |

| Source IP | Vector    | IF Score | Conf  | Priority | Phase   | Offences | Duration | Unblocked At        | Reason                                    |
|-----------|-----------|----------|-------|----------|---------|----------|----------|---------------------|-------------------------------------------|
| 10.0.0.18 | SYN Flood | 0.6471   | 90.9% | Critical | Phase 2 | 1        | 10s      | 2026-09-15 10:17:52 | Attack persisted - escalating to time ban |
| 10.0.0.17 | SYN Flood | 0.6423   | 92.1% | Critical | Phase 2 | 1        | 10s      | 2026-09-15 10:17:50 | Attack persisted - escalating to time ban |
| 10.0.0.16 | SYN Flood | 0.6363   | 92.5% | Critical | Phase 2 | 1        | 10s      | 2026-09-15 10:17:50 | Attack persisted - escalating to time ban |
| 10.0.0.6  | UDP Flood | 0.5891   | 92.4% | High     | Phase 2 | 1        | 1m 1s    | 2026-09-15 10:16:29 | Ban Expired                               |

**Verification and Approval**

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

\_\_\_\_\_  
SIGNATURE OVER PRINTED NAME  
**Network Administrator**

\_\_\_\_\_  
DATE